Digital Research Toolkit for Linguists

Week 12: Creating scientific documents with Quarto and plain text editors

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Psycholinguistics and Cognitive Modeling Lab

Tidyverse practive + `ggplot2`

22 exercises on `ggplot2`:
`https://anna-pryslopska.shinyapps.io/TidyversePractice`

17. Plotting

I class, we have been using the `ggplot2` and `esquisse` packages to visualize data. In this tutorial, you will practice plotting with the former package.

`ggplot2` was created by Hadley Wickham. Its approach to data viz is that of a layered grammar of graphics. You design and construct graphics in a structured manner from data upwards.

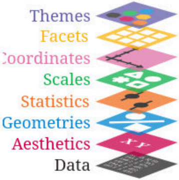

Diagram from slides in Week 7

Exercise 17.1: Create a ggplot object

Create a ggplot object from the `mtcars` data.

Questions?

Data	Understand	Communicate	Share
create, find, data types, formats, encoding	R, packages, import, clean, transform, debug, visualize, export	document, create clean and beautiful reports	connect, collaborate, backup, replicate

Table of contents

Documentation recap

Goals of documentation

Definition

Documentation is the process of recording information to describe, explain, or instruct.

Types

Internal Documentation: Code comments, inline explanations.
External Documentation: Manuals, READMEs, tutorials.

Why bother

Ensures your work is easy to understand, remember, reproduce, share, update, improve, and troubleshoot.

Key principles of documentation

Understandable	to your audience (at least yourself)
Short	people don’t read
Descriptive	what is x and what does it do
Focused	get to the point
Complete	data, packages, etc.
Up-to-date	what changed?
Timestamped	when did this happen?

Bonus: Use version control (e.g., Git)

Scientific document structure

What is a scientific document

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Brain and Cognition

journal homepage: www.elsevier.com/locate/b&c

Is there a difference between stripy journeys and stripy ladybirds? The N400 response to semantic and world-knowledge violations during sentence processing

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A R T I C L E I N F O

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Language comprehension
Semantic violations
World-knowledge violations
N400
EEG/ERP

A B S T R A C T

The distinction between linguistic and non-linguistic knowledge is particularly relevant because it is related to the principle of compositionality during sentence comprehension. Hagoort, Hald, Bastiaansen, and Petersson (2004) challenged the distinction between linguistic and non-linguistic knowledge. Here, we investigate how linguistic and non-linguistic violations are processed in a setting adapted from Hagoort et al., whilst in contrast to Hagoort, keeping the critical word identical. In line with the findings by Hagoort et al., our results showed largest N400 amplitudes for semantic violations ('Journeys are stripy'), followed by non-linguistic world-knowledge violations ('Ladybirds are stripy') and contingent sentences ('Trousers are stripy'), and finally by correct sentences ('Zebras are stripy'). Traditional fractional area and relative criterion measures of peak and onset latencies showed no effect of violation type. Interestingly, the semantic violation condition crossed a fixed criterion earlier than the word-knowledge violation condition. In conclusion, our data suggests that the question regarding the distinction between linguistic- and non-linguistic knowledge in terms of language integration remains open. Implications for future studies addressing the difference between linguistic and non-linguistic knowledge are discussed.

1. Introduction

Words are the building blocks of language. However, what makes human language so fascinating is the ability to express an unlimited number of thoughts and ideas. Language allows us to communicate about almost everything, for example, about things we have experienced in the past, things we would like to happen in the future, but also about things that will never happen or do not even exist in this world. In order to do so, we use various syntactic rules that combine the words into sentences. As these rules typically work on the sentence level, the sentence has often been regarded as the core unit of language production and comprehension. Semanticists widely agree that some version of Frege's (1892) principle of compositionality must hold (e.g. Pagin & Westerståhl, 2011), according to which the meaning of a complex expression is a function of the meanings of its parts and the ways they are syntactically combined. However, other sources of knowledge that do not purely derive from our linguistic knowledge about the meaning of words or syntactic structures are also accessed during language comprehension. One prominent source of this type of non-linguistic knowledge is our knowledge about the world (Bierwisch, 2007; Hobbs, 2011; Isberner & Richter, 2014; Jackendoff, 2002; Jackendoff, 2003; Lang & Maienborn, 2011; Marques, Canessa, & Cappa, 2009; Richter, Schroeder, & Wöhrmann, 2009). In the past decades a lively discussion has evolved regarding the question whether linguistic and non-linguistic sources of knowledge are accessed and integrated separately during language comprehension or whether they are processed in an all-in-one step.

Models that assume the principle of compositionality typically suggest that during meaning composition the semantics of an expression is derived purely on the basis of our linguistic knowledge (knowledge about word meaning, syntactic structure, etc.), and is thus separate from the sentence's truth-value with regard to our knowledge about the world (Hagoort & van Berkum, 2007). For example, a sentence such as *Ladybirds are stripy* allows building a coherent semantic interpretation and is unproblematic when only taking into account our linguistic knowledge about the words in the sentence and its syntactic structure. However, if we additionally take into account our knowledge about the world, then we must conclude that the sentence is not true, as ladybirds

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in our world typically are dotted and not stripy. In contrast, for a sentence such as *Journeys are stripy*, a violation is already indicated when only taking into account our linguistic knowledge. Specifically, the adjectival predicate 'stripy' demands an argument of type 'physical object', and thus cannot be combined with an event-denoting expression (i.e. *journey*) in order to derive a coherent semantic interpretation. That is, violation of a predicates's selectional restrictions leads to semantic anomaly, which is to be distinguished from falsity (world knowledge violation); see, e.g. Asher (2011).

One straightforward possibility to transfer the Fregean principle of compositionality into a processing account is thus to assume a two-step model of language comprehension, as outlined by Hagoort and van Berkum (2007): In a first step, the meaning of a sentence is constructed locally, by taking into account the meaning of the individual words and their syntactic combination. Other types of knowledge are integrated in a second step of processing. These other types of global pragmatic knowledge include the comprehender's general world-knowledge, knowledge resulting from the prior discourse and knowledge about the context of the utterance such as knowledge about the speaker's identity and gestures.

However, the empirical basis for such a model of sentence comprehension is rather controversial. Lattner and Friederici (2003) showed in an electrophysiological (EEG) study that non-linguistic information – specifically the information about a speaker's voice (male vs. female) and about gender stereotypes – was integrated rather late during sentence comprehension and occurred after the initial semantic analysis of a sentence had been completed. This finding seems to fit well with a two-step model of language comprehension. In contrast, recent studies question the assumptions regarding a separated integration of linguistic and non-linguistic knowledge during language understanding and instead suggest a one-step model of comprehension (for a review see: Hagoort & van Berkum, 2007). Indeed, manifold studies succeeded in showing that knowledge from various information sources are taken into account as early as sentence-based linguistic information during sentence comprehension. A key study by Hagoort et al. (2004) investigated whether the brain reflects differences between processing world-knowledge and semantic violations during sentence understanding. In order to differentiate between a one-step and a two-step model of sentence comprehension, EEG recordings were the central measurements in this study. EEG recordings typically allows one to measure real-time brain activity, within a resolution of milliseconds, and consequently are often used to investigate the precise time-course of cognitive processes (e.g., Dudschig & Jentzsch, 2008). In Hagoort et al.'s study, participants read sentences whose first conjunct expressed either a true statement such as *The Dutch trains are yellow (and very crowded)*, a world-knowledge violation such as *The Dutch trains are white (and very crowded)* or a semantic violation such as *The Dutch trains are sour (and very crowded)*. Thus, in order to create linguistic violations, the authors use the selection restriction criterion – whereby they violate the selection criterion that 'a predicate requires an argument whose semantic features match that of its predicate' (*p.* 439) – (see also Warren & McConnell, 2007; for alternatives, see, Pylkkänen, Oliveri, & Smart, 2009). For each sentence the N400 was measured relative to critical word onset (i.e. *yellow*, *white* and *sour*). The results showed that there was no difference in the N400 onset or peak latency between the semantic and the world-knowledge violations. Also the amplitude differences were rather small – however significant – when comparing the semantic and the world-knowledge violation conditions. The authors concluded that world-knowledge violations and semantic violations are integrated in parallel during language comprehension, in contrast to what is proposed by two-step-models of language comprehension. This study is often referenced as a milestone within psycholinguistic research by showing that empirically no difference between integrating linguistic and non-linguistic violations can be shown. In the following years, this work has been very influential and built the basis for many psychological theories of language understanding suggesting that all types of knowledge come down to one source (e.g., Filik, 2008; Matsuki et al., 2011; McRae & Matsuki, 2009; Nieuwland & Kuperberg, 2008). Despite challenges from linguistic research suggesting that there are serious grounds for the distinction between linguistic and non-linguistic knowledge, the N400 results from Hagoort and colleagues have been widely interpreted as indicating that there is no evidence from empirical investigations that suggest a processing or integration difference.

Given the importance of the results by Hagoort et al. (2004) for psycholinguistic research, we resume the question whether there are any indications that comprehenders integrate semantic knowledge and world knowledge in separate time steps during sentence comprehension. Specifically, we investigated the brain's responses to different violation types within sentences. As in the study by Hagoort and colleagues we used the N400-complex (Kutas & Hillyard, 1980) as our main measurement and implemented a conceptual replication of the original study with minor – but potentially important - changes in the experimental setup (for discussion on the value of such replication studies see: Schmidt, 2009). Surprisingly, it is still widely debated what mechanisms underlie the N400-complex (for a review see Lau, Phillips, & Poeppel, 2008). An *integration view* proposes that the N400 reflects semantic integration processes, whereby the N400 amplitude is determined by the ease with which the critical word can be integrated in the specific context (e.g., Brown & Hagoort, 1993). Alternatively, there is the *lexical view* of the N400 suggesting that the N400 reflects non-combinatorial processes and is determined only by the ease of access to long-term memory representations that are associated with one particular word in a specific context (for a review see Kutas & Federmeier, 2000). Independent of the exact mechanisms underlying the N400 effect, comprehensive work has investigated the specific parameters influencing the N400 complex. For example, the N400 is highly sensitive to word frequency (e.g., Kutas & Hillyard, 1984) and semantic distance (i.e. word co-occurrence; Van Petten, 2014). In the current study, we therefore used materials that allowed controlling for these issues. In line with Hagoort et al.'s study, we used correct sentences (*Zebras are stripy*), semantically violated sentences (*Journeys are stripy*), and sentences violated according to general world knowledge (*Ladybirds are stripy*). Importantly, in all our conditions, the critical words were kept identical. This allowed us to rule out word-based effects on the N400 complex and thus decreases the general N400 variance between the conditions, resulting in increased power to find between-condition differences (see Barber, Doñamayor, Kutas, & Münte, 2010; Davidson, Hanulikova, & Indefrey, 2012; Filik, Leuthold, Moxey, & Sanford, 2011; Hinojosa, Moreno, Casado, Muñoz, & Pozo, 2005). We also used an increased number of correct filler trials (50%) in order to avoid a strong bias of the participants in expecting a violated sentence and developing unnatural reading and comprehension strategies (see Kretzschmar, Bornkessel-Schlesewsky, & Schlesewsky, 2009). Additionally, we constructed contingent sentences – such as *Trousers are stripy* – containing a statement that is true for some members of the mentioned category but false for others, in order to avoid an instant and automated truth-value judgment triggered merely by the choice of the sentences used here. If there is a difference between the integration of linguistic and non-linguistic knowledge as predicted by two-step models of language comprehension, we expect differences in the N400 amplitudes and/or latencies between the semantic-violation and the world-knowledge violation condition. In contrast, if linguistic and

Scientific articles are, in essence, 75%+ documentation.

What is a scientific document

non-linguistic sources of knowledge are integrated simultaneously during comprehension, no such differences should be observed.

As mentioned above, the central conclusions drawn from Hagoort et al.'s (2004) paper are based on the timing of the N400 complex. In the Hagoort study the onset latencies were calculated by testing when the violation conditions started to differ from each other. The difference waveform between the semantic and the world-knowledge violation condition was calculated and it was tested at what point in time this difference waveform significantly started to deviate from zero. The present study aimed at investigating the difference between semantic and world-knowledge violations with three main differences to the original study. First, the critical word was kept identical, in order to reduce word-based variance that might hide subtle differences between violation conditions. Second, additional correct and contingent sentences were added to the material in order to avoid biased reading strategies. Finally, in addition to traditional methods, recently established methods for detecting significant clusters in the EEG signal and dealing with the multiple comparisons problem in biological data (Maris & Oostenveld, 2007) were used in order to draw conclusions regarding the integration of world- versus semantic knowledge during sentence comprehension

2. Method

2.1. Participants

Thirty right-handed participants were included in the reported analysis (M_{age} = 22.70, SD_{age} = 2.72, range: 19–27, 8 male). All participants were native German speakers with normal or corrected to normal vision. Overall 35 participants were initially tested with five participants being removed due to excessive EEG artifacts resulting in a limited number of trials remaining for data processing (<60% of the trials accepted after preprocessing the data).

2.2. Material

Overall 360 German sentences were used in this experiment. 160 critical sentences and 180 filler sentences were constructed. Additionally 20 sentences were constructed for the practice trials. The critical sentences were divided into four subgroups. 40 sentences were correct sentences – true condition (e.g. '*Zebras are stripy*'). 40 sentences were constructed that were true for some members of the mentioned category – contingent condition (e.g. '*Trousers are stripy*'). 40 sentences were semantically violated – semantic condition (e.g. '*Journeys are stripy*'). Finally, another set of 40 sentences contained a world-knowledge violation – world-knowledge condition (e.g. '*Ladybirds are stripy*'). All sentences consisted of a noun, the copula *sein* ('to be') and an adjective (see Appendix, Table 1 for example sentences). The filler sentences were constructed similarly to the critical sentences and contained only true statements (e.g., '*Stones are hard*') in order to minimize the violation/true ratio in the experiment. The words were displayed in black on a light gray background. The nouns used in the experimental sentences were controlled for frequency with the Leipziger Wortschatzportal, $F(3,156) = 1.74$, $p = .16$ ($M_{Correct}$ = 1895, $M_{Contingent}$ = 5364, $M_{Semantic}$ = 3392, $M_{WorldKowledge}$ = 2787). The critical words were the adjectives; these were identical across all four conditions and each participant saw each adjective once in each condition. As a measure of word association between the noun and the adjective of each sentence, a latent semantic analysis (LSA; Landauer, Foltz, & Laham, 1998) was conducted using the LSAfun R-package (Günther, Dudschig, & Kaup, 2014) and previously established semantic spaces based on lemmatized newspaper and blog corpora (Günther, Dudschig, & Kaup, 2015). There was no main effect of LSA similarity between the four conditions, $F(3,156) = 1.44$, $p = .23$ ($M_{Correct}$ = 0.08, $M_{Contingent}$ = 0.06, $M_{Semantic}$ = 0.04, $M_{WorldKowledge}$ = 0.06). Given the rather small associations resulting from the LSA analyses an additional word association or predictability test was conducted. Specifically, as a type of cloze measurement, a sentence completion was conducted with twelve naïve German native speakers (M_{age} = 24.42, 10 female, all right-handed) that did not take part in the actual experiment. Participants were provided with the beginning of the sentence and had to provide five adjectives that came to their mind to end the sentence (Block & Baldwin, 2010; Bloom & Fischler, 1980; Taylor, 1953). If participants provided the correct adjective – according to our experimental sentences – this was counted as a hit, if they provided non-matching adjectives this was counted as a miss. Subsequently, the percentage of hits was calculated for each sentence. An ANOVA was conducted showing a clear effect of experimental condition in this cloze-measurement, $F(3,156) = 61.96$, $p < .001$. Between conditions comparison *t*-tests showed that the correct sentences were more often completed with the according final word than contingent sentences, $t(78) = 5.60$, $p < .001$ ($M_{Correct}$ = 59.79%, $M_{Contingent}$ = 19.79%). The contingent condition also differed from the world-knowledge ($M_{WorldKnowledge}$ = 0.21%), $t(78) = 4.32$, $p < .001$, and the semantic violation condition ($M_{Semantic}$ = 0.00%), $t(78) = 4.38$, $p < .001$). Most importantly, the semantic and the world knowledge condition did not differ significantly, $t(78) = -1.00$, $p = .32$.

2.3. Procedure and design

The experiment took place in a sound-attenuated booth. The experimental procedure was controlled by E-Prime (Psychology Software Tools, Inc., Version E-Prime 2.0.10.92; Schneider, Eschman, & Zuccolotto, 2002). Each trial started with the presentation of a fixation-cross displayed in the center of the screen for 1500 ms. The words were presented successively in the center of the screen. Each word was displayed for 300 ms followed by a 300 ms blank screen. The last word was followed by a 1000 ms blank screen. In approximately 8% of the trials, comprehension questions were asked. Participants had to indicate with a button press (n or v key) whether a statement (e.g., '*Stones are solid material*') is correct or not according to the preceding sentence (e.g. '*Stones are hard*'). These trials were added to keep the participants' attention to the sentences. The experiment consisted of one practice block and four experimental blocks. Each experimental block contained 40 critical sentences, ten of each condition, and 45 filler sentences. Thereby it was ensured that critical words did not repeat within a block, in order to avoid direct word repetitions and that each condition was equally likely to be presented by the first, the second, the third or the fourth instance of the critical word. The trial order within each block was randomized for each participant. Block order was varied between participants (version 1: 1-2-3-4, version 2: 4-3-2-1, version 3: 3-1-4-2, version 4: 2-4-1-3).

3. Electrophysiological measures

3.1. EEG recording

EEG activity was recorded continuously from 70 electrodes using a BioSemi EEG system. The location of the electrodes are as follows: midline electrodes (Fpz, AFz, Fz, FCz, Cz, CPz, Pz, POz, Oz, and Iz); left hemisphere electrodes (IO1, Fp1, AF3, AF7, F1, F3, F5, F7, F9, FC1, FC3, FC5, FT7, C1, C3, C5, M1, T7, CP1, CP3, CP5, TP7, P1, P3, P5, P7, P9, PO3, PO7, O1) and the homologue electrodes in the right hemisphere. Two additional electrodes (Common Mode Sense [CMS] active electrode and Drive Right Leg [DRL] passive electrode) were used as reference and ground electrodes, respectively (cf., www.biosemi/faq/cms&drl.htm). Vertical electroocular

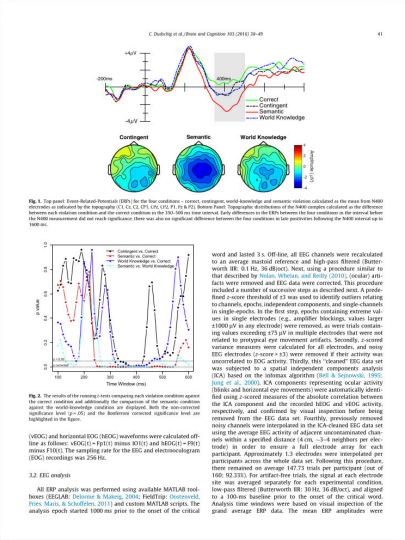


Fig. 1. Top panel: Event-Related-Potentials (ERPs) for the four conditions – correct, contingent, world-knowledge and semantic violation calculated as the mean from N400 electrodes as indicated by the topography (C1, Cz, C2, CP1, CPz, CP2, P1, Pz & P2). Bottom Panel: Topographic distributions of the N400 complex calculated as the difference between each violation condition and the correct condition in the 350–500 ms time interval. Early differences in the ERPs between the four conditions in the interval before the N400 measurement did not reach significance, there was also no significant difference between the four conditions in late positivities following the N400 interval up to 1600 ms.

Fig. 2. The results of the running *t*-tests comparing each violation condition against the correct condition and additionally the comparison of the semantic condition against the world-knowledge condition are displayed. Both the non-corrected significance level (p = .05) and the Bonferroni corrected significance level are highlighted in the figure.

(vEOG) and horizontal EOG (hEOG) waveforms were calculated off-line as follows: vEOG(t) = Fp1(t) minus IO1(t) and hEOG(t) = F9(t) minus F10(t). The sampling rate for the EEG and electrooculogram (EOG) recordings was 256 Hz.

3.2. EEG analysis

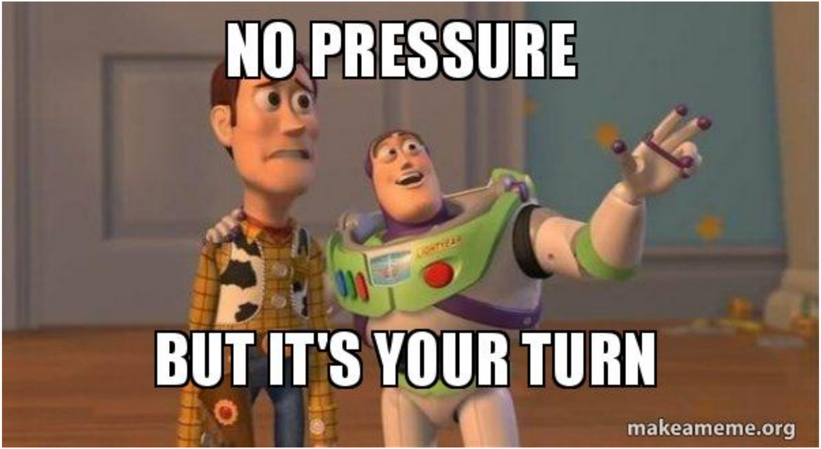
All ERP analysis was performed using available MATLAB toolboxes (EEGLAB: Delorme & Makeig, 2004; FieldTrip: Oostenveld, Fries, Maris, & Schoffelen, 2011) and custom MATLAB scripts. The analysis epoch started 1000 ms prior to the onset of the critical word and lasted 3 s. Off-line, all EEG channels were recalculated to an average mastoid reference and high-pass filtered (Butterworth IIR: 0.1 Hz, 36 dB/oct). Next, using a procedure similar to that described by Nolan, Whelan, and Reilly (2010), (ocular) artifacts were removed and EEG data were corrected. This procedure included a number of successive steps as described next. A predefined *z*-score threshold of ±3 was used to identify outliers relating to channels, epochs, independent components, and single-channels in single-epochs. In the first step, epochs containing extreme values in single electrodes (e.g., amplifier blockings, values larger ±1000 µV in any electrode) were removed, as were trials containing values exceeding ±75 µV in multiple electrodes that were not related to protypical eye movement artifacts. Secondly, *z*-scored variance measures were calculated for all electrodes, and noisy EEG electrodes (*z*-score > ±3) were removed if their activity was uncorrelated to EOG activity. Thirdly, this "cleaned" EEG data set was subjected to a spatial independent components analysis (ICA) based on the infomax algorithm (Bell & Sejnowski, 1995; Jung et al., 2000). ICA components representing ocular activity (blinks and horizontal eye movements) were automatically identified using *z*-scored measures of the absolute correlation between the ICA component and the recorded hEOG and vEOG activity, respectively, and confirmed by visual inspection before being removed from the EEG data set. Fourthly, previously removed noisy channels were interpolated in the ICA-cleaned EEG data set using the average EEG activity of adjacent uncontaminated channels within a specified distance (4 cm, ~3–4 neighbors per electrode) in order to ensure a full electrode array for each participant. Approximately 1.3 electrodes were interpolated per participants across the whole data set. Following this procedure, there remained on average 147.73 trials per participant (out of 160; 92.33%). For artifact-free trials, the signal at each electrode site was averaged separately for each experimental condition, low-pass filtered (Butterworth IIR: 30 Hz, 36 dB/oct), and aligned to a 100-ms baseline prior to the onset of the critical word. Analysis time windows were based on visual inspection of the grand average ERP data. The mean ERP amplitudes were

Journal article, (text)book, report, case study, manual, proceedings, poster, presentation, opinion piece, thesis, … → same in industry!

Typical scientific document

Title	witty or not, but informative
Abstract	elevator pitch
Introduction	topic, background, motivation, hypothesis, aim, …
Methods	description of how the research was conducted
Analysis	data analysis info
Results	study findings
Discussion	interpretation of the results
References	works cited
Appendices	supplementary material

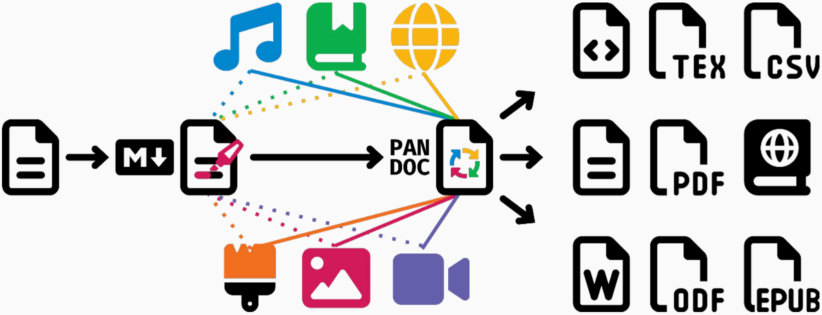
Practice makes perfect



In the homework, **YOU** get to write a scientific report on the Moses illusion experiment!(But only ~20 sentences for now)

Scientific documents with Quarto

Quarto recap



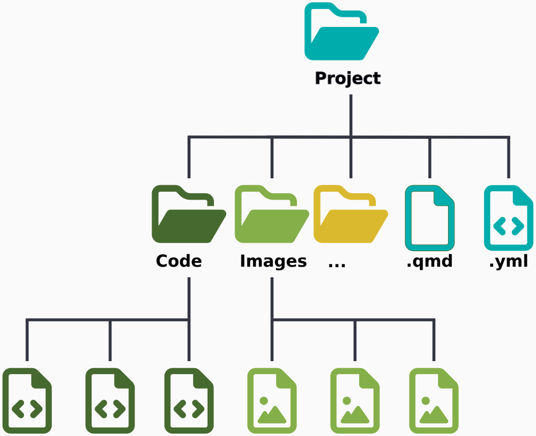
Quarto uses pandoc markdown with extra syntax and features to create dynamic documents, presentations, and websites.It supports many programming languages/formats and can integrate code, formatted text, and visualizations.

Scientific report with Quarto

One main file to rule them all: `index.qmd`

(Semi-)optional configuration file: `_quarto.yaml`

Supplementary files: images, code, bibliography, ...



YAML header

YAML (aka “Yet Another Markup Language”, or “YAML ain’t markup language”) metadata consist of key-value pairs that define core document settings:

```yaml
---
title: My awesome title
author:
  - name: Name Surname
    corresponding: true
    email: email@email.com
    affiliations:
      - University of Awesome
keywords:
  - something
  - something something
abstract: |
  Read my awesome paper
date: last-modified
number-sections: true
---
```

Global output options

You can specify global output options (= what you usually want to happen) and then overwrite them when needed.

The following code is included within the YAML header:

- `echo` = include the code itself in the output
- `include` = include the **results** of the code in the output
- `warning` = show warning messages from the code

```
---
execute:
  echo: false
  include: true
  warning: false
---
```

Rest of the f///ing text

```
## Section 1
But I **must** explain to you how this mistaken idea
of denouncing pleasure and praising pain was born
and I will give you a complete account of the system,
and expound the actual teachings of the great
explorer of the truth, the master-builder of human
happiness.

![Happiness](happiness.png)

### Subsection 1

No one rejects, dislikes, or avoids pleasure itself,
because it is pleasure, but because those who do
not know how to pursue pleasure rationally encounter
consequences that are extremely painful.
```

Tables

````
| Col1 | Col2 | Col3 |
|------|------|------|
|      |      |      |
|      |      |      |

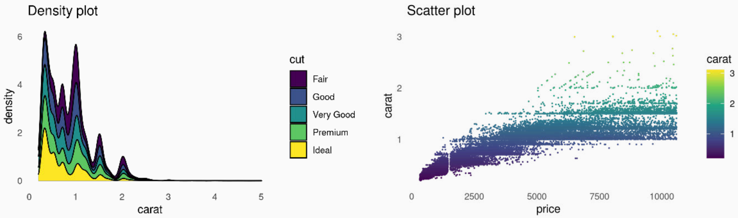
: Caption here {#tbl-id}

```{r}
#| label: tbl-id
#| tbl-colwidths: [60,40]
#| tbl-cap: Table caption here
print(table)
```
````

Now I can cross-reference `@tbl-id` in the text. Or make a manual reference: `[Table reference](#tbl-id)`. It will show up as Table 1 and Table reference, respectively.

Figures

```
```{r}
#| label: fig-id
#| layout-ncol: 2
#| fig-cap: Figure caption here
print(plot1)
print(plot2)
```
```



Now I can cross-reference `@fig-id` in the text.

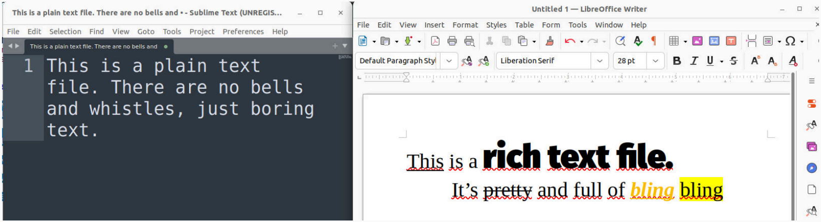
Plain text editors

Text editors

allow you to write, open, edit, and create plain text files.

Plain vs. rich text

No formatting (typeface, size, weight, style etc.), nothing unexpected. Only text.



What is a text editor?

- Most portable way of sharing files.
- Useful for previewing data or writing code.
- Highlight any syntax (e.g. R, markdown, $\TeX$)
- You can open any file in a text editor.
- Usually, nothing unexpected will happen.

What good is a plain text file?

Simplicity

No learning curve, minimalist interface, no distractions

Performance

Lightweight, fast even with large files, easy on the computer

Portability

Text files are readable across operating systems, compatible with everything, virtually the same since 1950s

Customization

Plugins and scripts can change an editor to a personal assistant

What good is a plain text file?

No Lock-In

No proprietary formats, independent, can be edited with any editor

Version Control Friendly

Easy to track changes and merge

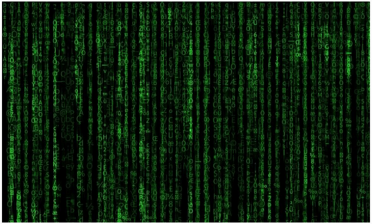
Security

Lower risk of viruses and embedded malicious code, unlikely to be targets

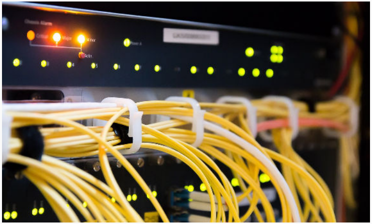
Focus on Content

Focus on writing and coding without formatting distractions, encourages good practices and clear documentation

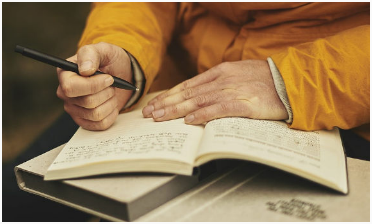
What use is a plain text file?



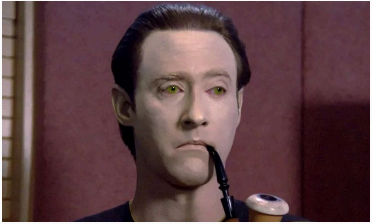
Writing and editing code.



System administration.



Creating quick notes, drafts, documentation.



Writing, manipulating, processing text data.

Which editor to chose?

Opinions on text editors are like butts: everyone has one.

Which do I have to install?

1. **Sublime text** `https://www.sublimetext.com/`
   beautiful, simple, free but will nag you to buy it
2. Microsoft VS Code `https://code.visualstudio.com/`
   good for coding, git integration, overkill
3. **Notepad++** `https://notepad-plus-plus.org/`
   step up from plain Notepad, good for beginners
4. Honorable mentions for more overkill:
   - **Emacs** `https://www.gnu.org/software/emacs/`
   - **Vim** `https://www.vim.org/`

Questions?

Summary

- ✔ documentation recap
- ✔ scientific document structure
- ✔ writing scientific documents in Quarto
- ✔ including tables and plots
- ✔ cross-referencing and hyperlinking
- ✔ plain text editors
- ⏩ literature research, bibliography management and large projects

Homework assignment

Homework assignment due June 30 at 12:00

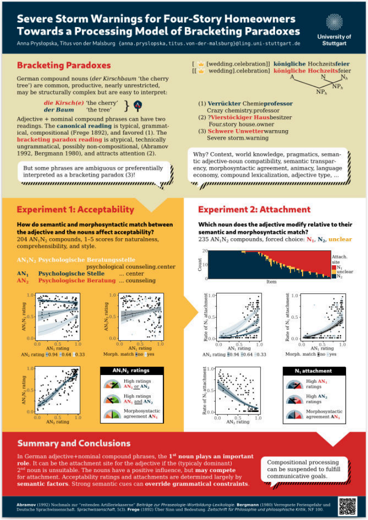
- Complete assignment 10 ($\rightarrow$ ILIAS)
- File: `index.qmd`
- Do not use AI.

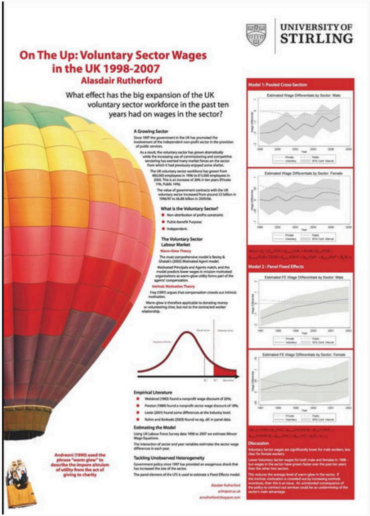
Poster and poster presentation

Imagine you're presenting your outstanding thesis at a scientific conference.

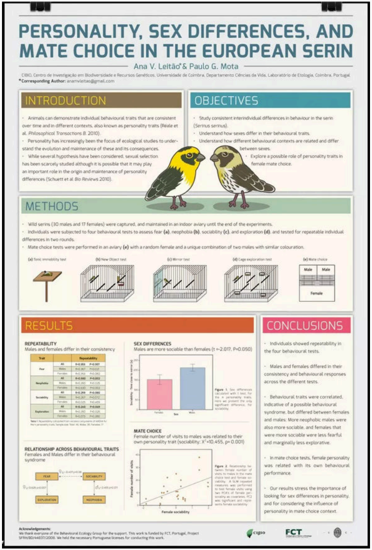
- Topic: Moses illusion with NEW data
- A0 size
- Landscape or portrait orientation
- Scientific publication structure with plots and tables
- Following the POUR principles
- PDF file (**do NOT print it**)
- Created in any software you want: OpenOffice Impress, LaTeX, PowerPoint, Keynote, Canva, hand-drawn, Quarto, Photoshop, InDesign, Inkscape, …
- Due before 2026, but for 9 CP you need to give me a 5–10 minute "presentation" walking me through the poster

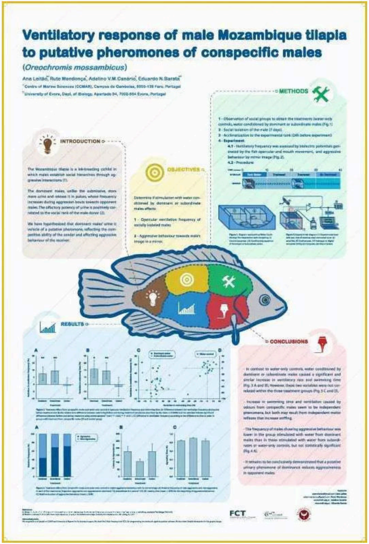
Poster and poster inspiration



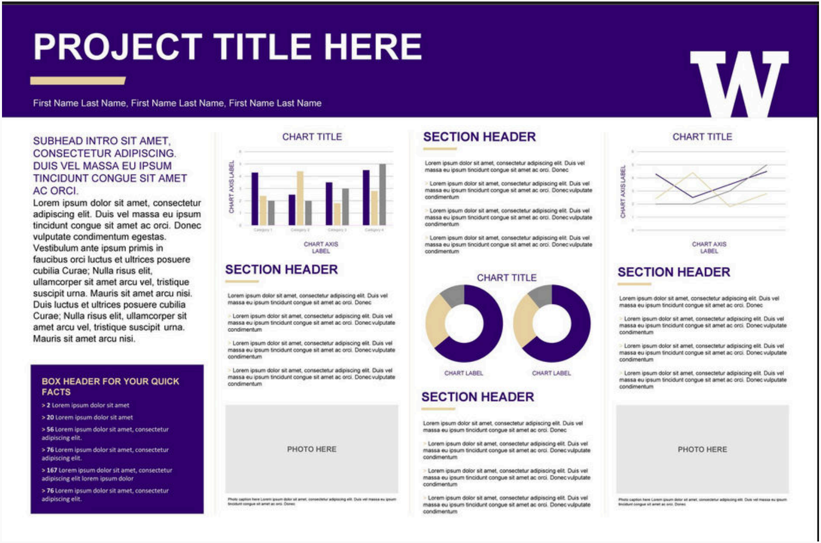


Poster and poster inspiration

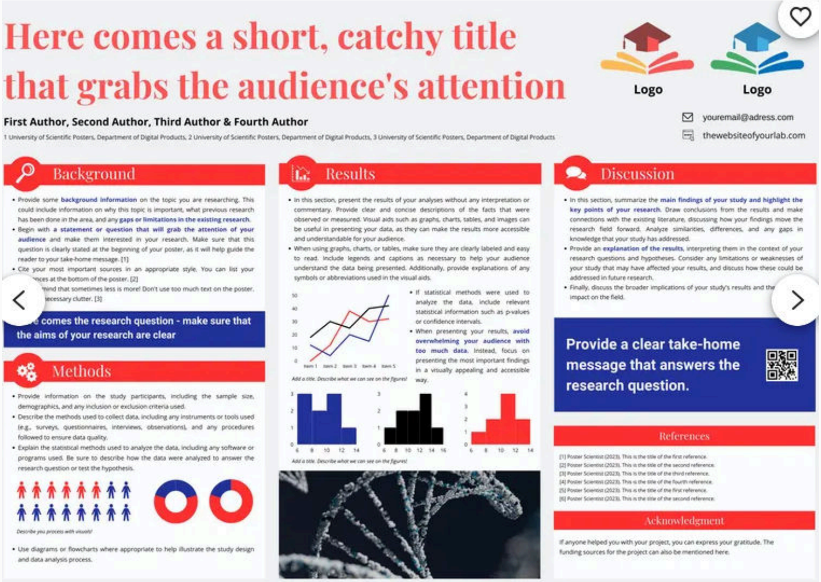




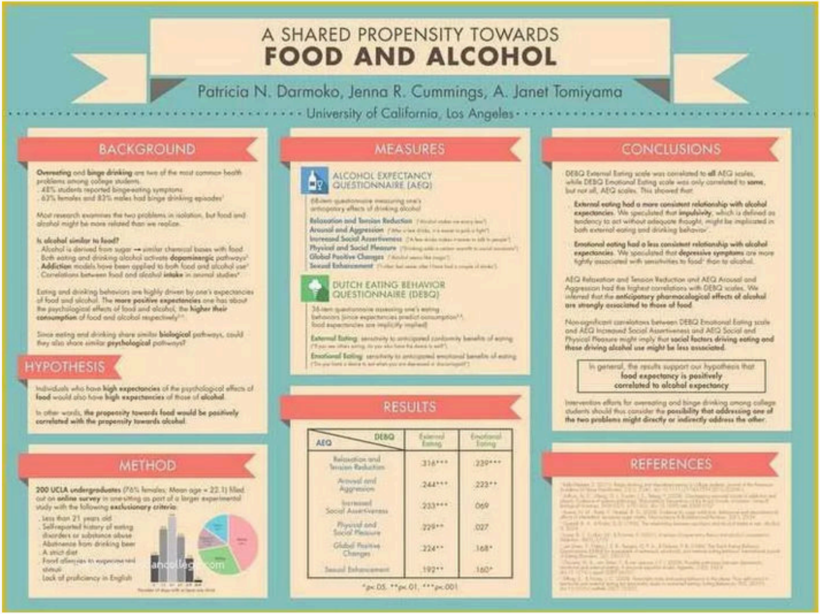
Poster and poster inspiration



Poster and poster inspiration



Poster and poster inspiration



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